

First I ran both codes and did a dump to make sure everything was connected properly:

```
karinajadia@Ubuntu-2004:~$ sudo ~/pox/pox.py misc.practice_controller
POX 0.7.0 (gar) / Copyright 2011-2020 James McCauley, et al.
WARNING:version:Support for Python 3 is experimental.
INFO:core:POX 0.7.0 (gar) is up.
```

All of the hosts are connected to the correct IP address that I had configured, so the dump was successful!

```
mininet> pingall
*** Ping: testing ping reachability
h1 -> h2 h3 x h5
h2 -> h1 h3 x h5
h3 -> h1 h2 x h5
h4 -> x x x x
h5 -> h1 h2 h3 x
*** Results: 40% dropped (12/20 received)
```

Here are the iperfs to see how each host interacts with each other. Unfortunately my h4 cannot send any traffic to the other hosts, when I try to run iperf it stalls forever and simply doesn't work. I have all the rules and flow table set up in my controller so I don't understand why it doesn't work. The rest of the hosts can send data to each other.

I also did dpctl dump-flows to show how traffic moves between the hosts after a pingall.

```
mininet> dpctl dump-flows
*** s1 -----
cookie=0x0, duration=17.376s, table=0, n_packets=1, n_bytes=98, idle_timeout=30, hard_timeout=30, icmp,vlan_tci=0x0000,d_lsrc=00:00:00:00:00:05,d_l_dst=00:00:00:00:00:01,nw_src=10.5.5.50,nw_dst=10.1.1.10,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s1-eth1"
cookie=0x0, duration=17.375s, table=0, n_packets=1, n_bytes=98, idle_timeout=30, hard_timeout=30, icmp,vlan_tci=0x0000,d_lsrc=00:00:00:00:00:01,d_l_dst=00:00:00:00:00:05,nw_src=10.1.1.10,nw_dst=10.5.5.50,nw_tos=0,icmp_type=0,icmp_code=0 actions=output:"s1-eth2"
*** s2 -----
cookie=0x0, duration=17.364s, table=0, n_packets=1, n_bytes=98, idle_timeout=30, hard_timeout=30, icmp,vlan_tci=0x0000,d_lsrc=00:00:00:00:00:05,d_l_dst=00:00:00:00:00:02,nw_src=10.5.5.50,nw_dst=10.2.2.20,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s2-eth1"
cookie=0x0, duration=17.363s, table=0, n_packets=1, n_bytes=98, idle_timeout=30, hard_timeout=30, icmp,vlan_tci=0x0000,d_lsrc=00:00:00:00:00:02,d_l_dst=00:00:00:00:00:05,nw_src=10.2.2.20,nw_dst=10.5.5.50,nw_tos=0,icmp_type=0,icmp_code=0 actions=output:"s2-eth2"
```

```
*** s3 -----
cookie=0x0, duration=17.365s, table=0, n_packets=1, n_bytes=98, idle_timeout=30, hard_timeout=30, icmp,vlan_tci=0x0000,d_lsrc=00:00:00:00:00:05,d_l_dst=00:00:00:00:00:03,nw_src=10.5.5.50,nw_dst=10.3.3.30,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s3-eth1"
cookie=0x0, duration=17.364s, table=0, n_packets=1, n_bytes=98, idle_timeout=30, hard_timeout=30, icmp,vlan_tci=0x0000,d_lsrc=00:00:00:00:00:03,d_l_dst=00:00:00:00:00:05,nw_src=10.3.3.30,nw_dst=10.5.5.50,nw_tos=0,icmp_type=0,icmp_code=0 actions=output:"s3-eth2"
*** s4 -----
cookie=0x0, duration=27.414s, table=0, n_packets=0, n_bytes=0, idle_timeout=30, hard_timeout=30, icmp,vlan_tci=0x0000,d_lsrc=00:00:00:00:00:05,d_l_dst=00:00:00:00:00:04,nw_src=123.45.67.89,nw_dst=10.5.5.50,nw_tos=0,icmp_type=8,icmp_code=0 actions=drop
cookie=0x0, duration=17.351s, table=0, n_packets=1, n_bytes=98, idle_timeout=30, hard_timeout=30, icmp,vlan_tci=0x0000,d_lsrc=00:00:00:00:00:05,d_l_dst=00:00:00:00:00:04,nw_src=10.5.5.50,nw_dst=123.45.67.89,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s4-eth2"
```

```
karinajadia@Ubuntu-2004:~$ sudo python3 ~/project3.py
[sudo] password for karinajadia:
Delete me!
mininet> dump
<Host h1: h1-eth0:10.1.1.10 pid=4582>
<Host h2: h2-eth0:10.2.2.20 pid=4584>
<Host h3: h3-eth0:10.3.3.30 pid=4586>
<Host h4: h4-eth0:123.45.67.89 pid=4588>
<Host h5: h5-eth0:10.5.5.50 pid=4590>
<OVSSwitch s1: lo:127.0.0.1,s1-eth1:None,s1-eth2:None pid=4595>
<OVSSwitch s2: lo:127.0.0.1,s2-eth1:None,s2-eth2:None pid=4598>
<OVSSwitch s3: lo:127.0.0.1,s3-eth1:None,s3-eth2:None pid=4601>
<OVSSwitch s4: lo:127.0.0.1,s4-eth1:None,s4-eth2:None pid=4604>
<OVSSwitch s5: lo:127.0.0.1,s5-eth1:None,s5-eth2:None,s5-eth3:None,s5-eth4:None,s5-eth5:None pid=4607>
<RemoteController c0: 127.0.0.1:6633 pid=4576>
mininet>
```

My pingall also showed the hosts connecting to each other. h4 cannot send ICMP packets to anyone because they are untrusted. All of the other hosts can ping each other.

```
mininet> iperf h1 h2
*** Iperf: testing TCP bandwidth between h1 and h2
*** Results: ['27.3 Gbits/sec', '27.2 Gbits/sec']
mininet> iperf h1 h3
*** Iperf: testing TCP bandwidth between h1 and h3
*** Results: ['30.4 Gbits/sec', '30.5 Gbits/sec']
mininet> iperf h1 h5
*** Iperf: testing TCP bandwidth between h1 and h5
*** Results: ['27.9 Gbits/sec', '28.1 Gbits/sec']
mininet> iperf h2 h3
*** Iperf: testing TCP bandwidth between h2 and h3
*** Results: ['31.0 Gbits/sec', '30.9 Gbits/sec']
mininet> iperf h2 h5
*** Iperf: testing TCP bandwidth between h2 and h5
*** Results: ['31.9 Gbits/sec', '32.0 Gbits/sec']
mininet> iperf h3 h5
*** Iperf: testing TCP bandwidth between h3 and h5
*** Results: ['28.4 Gbits/sec', '28.6 Gbits/sec']
```

```
*** s5 -----
cookie=0x0, duration=17.418s, table=0, n_packets=1, n_bytes=98, idle_timeout=30, hard_timeout=30, icmp,vlan_tci=0x0000,d_lsrc=00:00:00:00:00:05,d_l_dst=00:00:00:00:00:01,nw_src=10.5.5.50,nw_dst=10.1.1.10,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s5-eth2"
cookie=0x0, duration=17.414s, table=0, n_packets=1, n_bytes=98, idle_timeout=30, hard_timeout=30, icmp,vlan_tci=0x0000,d_lsrc=00:00:00:00:00:01,d_l_dst=00:00:00:00:00:05,nw_src=10.1.1.10,nw_dst=10.5.5.50,nw_tos=0,icmp_type=0,icmp_code=0 actions=output:"s5-eth1"
cookie=0x0, duration=17.398s, table=0, n_packets=1, n_bytes=98, idle_timeout=30, hard_timeout=30, icmp,vlan_tci=0x0000,d_lsrc=00:00:00:00:00:05,d_l_dst=00:00:00:00:00:03,nw_src=10.5.5.50,nw_dst=10.2.2.20,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s5-eth3"
cookie=0x0, duration=17.396s, table=0, n_packets=1, n_bytes=98, idle_timeout=30, hard_timeout=30, icmp,vlan_tci=0x0000,d_lsrc=00:00:00:00:00:02,d_l_dst=00:00:00:00:00:05,nw_src=10.2.2.20,nw_dst=10.5.5.50,nw_tos=0,icmp_type=0,icmp_code=0 actions=output:"s5-eth1"
cookie=0x0, duration=17.386s, table=0, n_packets=1, n_bytes=98, idle_timeout=30, hard_timeout=30, icmp,vlan_tci=0x0000,d_lsrc=00:00:00:00:00:03,d_l_dst=00:00:00:00:00:05,nw_src=10.5.5.50,nw_dst=10.3.3.30,nw_tos=0,icmp_type=8,icmp_code=0 actions=output:"s5-eth4"
cookie=0x0, duration=17.384s, table=0, n_packets=1, n_bytes=98, idle_timeout=30, hard_timeout=30, icmp,vlan_tci=0x0000,d_lsrc=00:00:00:00:00:03,d_l_dst=00:00:00:00:00:05,nw_src=10.3.3.30,nw_dst=10.5.5.50,nw_tos=0,icmp_type=0,icmp_code=0
```